DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #01H: PATHS & SUBPATHS (PART 8 — GRANULAR AUDIT OF GRAND DAO WAVE 2B: `DIVINE_SUMMONER`, `DIVINE_POSSESSION`, `DIVINE_CHANNELER`, `OCCULT_MAGE`, `VOID_HERETIC_PACT`, `DEVOURING_FLESH`, `NETHER_GRAVE` & ALL 28 SUBPATHS — COMPLETING ALL 43 PATHS & 286 SUBPATHS)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Series Completion (`01A–01H` Complete 43-Path / 286-Subpath Architecture & Granular Audit)**:
  - `01A–01B`: Shared Macro Lifecycle, Path Library, Retraining & Stage Equivalence Math.
  - `01C`: Group 1 — 5 Founding Pillars (`65` Subpaths).
  - `01D`: Group 2 — 7 Specialist Expansion Paths (`67` Subpaths).
  - `01E`: Group 3 — 8 Cosmic, Spacetime, Weather, Dharma & Geomancy Paths (`48` Subpaths).
  - `01F`: Group 4 — 8 Mercantile, Literati, Archivist, Wandering, Sonic, Venom, Puppetry & Shifter Paths (`46` Subpaths).
  - `01G`: Group 5 — 8 Samurai, Shinobi, Arena, Gambler, Voyager, Husbandry & Ancestral Paths (`32` Subpaths).
  - **`01H` (This Final Path Document)**: Group 6 — **7 Higher-Realm Pantheon, Southeast-Asian Occult, Eldritch Pact, Flesh-Devouring & Necropolis Grand Dao Paths (`DIVINE_SUMMONER` [4], `DIVINE_POSSESSION` [4], `DIVINE_CHANNELER` [4], `OCCULT_MAGE` [4], `VOID_HERETIC_PACT` [4], `DEVOURING_FLESH` [4], `NETHER_GRAVE` [4] — `28` Subpaths; `65+67+48+46+32+28 = 286/286` Subpaths Audited!)**.

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`).
2. **40-Script Load Order & Strict Closed-Loop Conservation**: Flesh devouring, shrine consecrations, and demonic/eldritch pacts must consume **personal** or **authorized** resources (`Debit === Credit`) and never hijack unrelated NPCs via `|| aliveCults[0]` fallbacks.
3. **100% Emergent Cultural & Moral Coherence**: Orthodox sects do not casually run Demonic Blood Shrines; Flesh Devourers eat anatomy-matched ores/beast parts/organs rather than raw rice or whetstones; Corpse Explosions require actual slain corpses on the battlefield!

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1. `[PATH-PANTHEON-01]` Paths #31, #32 & #33: Higher-Realms Divine Triad (`DIVINE_SUMMONER` [4], `DIVINE_POSSESSION` [4], `DIVINE_CHANNELER` [4] — 12 Subpaths)

1.1 CRITICAL P0 MATH BUG: `deity.atkMult - 1` Wipes `-68%` to `-82%` of Stats (`js/cultivation.js` & `js/stats.js`)!

- **Origin of Regression**:
  - In `v2.2.0`, all 20 Deities in `CULT.HIGHER_REALMS_PANTHEON.DEITIES` (across 5 Higher Civilizations: `HEAVENLY_COURT_DAO`, `VEDIC_VAJRA_REALM`, `DUAT_NETHER_SANDS`, `OLYMPUS_TARTARUS`, `HIMAVANTA_SIAM_DEVA`) defined `atkMult`, `matkMult`, and `defMult` as **additive bonus fractions (`0.18` to `0.32`, meaning `+18%` to `+32%`)**!
  - In `v2.4.0` (`NEW-03`), when wiring `getUnitBlessingBonuses(n)` into `stats.js:396, 591`, the code assumed `atkMult` was `1.24` and wrote **`deity.atkMult - 1`** (`0.24 - 1 = -0.76`)! (Same bug check needed on `DEMONIC_PANTHEON` if any patron uses `< 0.8` additive notation!).
- **Catastrophic Runtime Effect in `JS2-v2.6.0`**:
  - Any `DIVINE_SUMMONER`, `DIVINE_POSSESSION`, or `DIVINE_CHANNELER` disciple who spends Spirit Stones to form a Pact with Pangu, Shiva, Ra, Anubis, Zeus, or Phra Phirap **immediately loses `-68%` to `-82%` of their ATK, MATK, and DEF**!
- **Exact Fix**:

```javascript
const toBonusPct = v => (v == null ? 0 : (v > 0.8 ? v - 1 : v));
```

Scale the resulting positive bonus by `cosmicScale` (`0.45 + 0.55 * stageProgress`, honoring the `10^-12` $\to$ `5%` Non-Linear Cosmic Fraction lore curve without overflowing combat math!).

1.2 Orphaned `SECT.COSMIC_SHRINE.consecrateOffering` & Subpath-to-Pantheon Matching

- **Verified Defect (`N-03`)**: `SECT.COSMIC_SHRINE.consecrateOffering` had zero autonomous callers outside `v220` tests (or hijacked `aliveCults[0]` in monthly crons).
- **Authentic Subpath Deity Selection**: When a `DIVINE_*` cultivator reaches Stage 2+ with personal stones/offerings, they should autonomously walk to `shrine` (`AI.ACTS.consecrateDeityPact`) and form a pact **matching their Subpath's Mythology**:
  - `SOLAR_AVATAR_VESSEL` $\to$ Ra / Donghuang Taiyi (`FIRE` / `LIGHT_YANG`)
  - `STORM_WRATH_MEDIUM` / `THUNDERBOLT_WIELDER` $\to$ Zeus / Indra (`LIGHTNING` / `WIND`)
  - `VAJRA_EARTH_ANCHOR` $\to$ Pangu / Nuwa (`EARTH_METAL`)
  - `MAHADEVA_DESTROYER` / `THIRD_EYE_INSIGHT` $\to$ Shiva Nataraja / Phra Phirap / Erlang Shen
  - `JUDGMENT_SCALE_BORROWER` / `NETHER_GATE_CALLER` $\to$ Anubis / Hades / Yama
  - `SERPENT_GRACE_BORROWER` $\to$ Naga Sovereign / Phra Mae Thorani (`WATER_ICE` / `WOOD_LIFE`)

1.3 Differentiating the Three Divine Conduit Paths Mechanically

Currently all three paths behave like generic `divineResonance` stat sticks. Enforce their distinct 3-way architecture:

1. **`DIVINE_SUMMONER` (อัญเชิญเงาทวยเทพลงสนาม — `CELESTIAL_BEAST_HERALD`, `WAR_DEITY_CONVOKER`, `NETHER_GATE_CALLER`, `PANTHEON_CHOIR_CONDUCTOR`)**:
   - Spawns an **external Deity Avatar Projection** (`isSummon: true`, `hpMax` & offense scaled by `cosmicFractionForStage`), keeping the summoner in `BACK_ROW`; if interrupted (`stun`/`silence`) during gate-opening windup, suffers **Dimensional Rift Recoil** (`-10% HP`).
2. **`DIVINE_POSSESSION` (ร่างทรงเทวะประทับร่าง — `SOLAR_AVATAR_VESSEL`, `STORM_WRATH_MEDIUM`, `VAJRA_EARTH_ANCHOR`, `MAHADEVA_DESTROYER`)**:
   - **Zero external summon**; instead enters **Avatar Descent Trance** (`18` ticks: `+32% MaxHP` temporary shield, `+28% ATK/MATK`, `FRONT_ROW` poise & `fear`

immunity), followed by **Post-Trance Vessel Strain (อ่อนล้าหลังถอนทรง: `-18 energy` & 4-tick `exposed` cooldown window)**!

3. **`DIVINE_CHANNELER` (ยืมกฎสวรรค์เคลือบศาสตรา — `JUDGMENT_SCALE_BORROWER`, `THUNDERBOLT_WIELDER`, `THIRD_EYE_INSIGHT`, `SERPENT_GRACE_BORROWER`)**:
    - **Zero summon & zero trance exhaustion**; coats equipped weapon with **Divine Law Passives**:
        - `JUDGMENT_SCALE_BORROWER`: `+2%` bonus true judgment damage per `10` target `sinKarma` / kills (capped `+30%`).
        - `THIRD_EYE_INSIGHT`: Grants **True Sight**—completely ignores target `shadowClone` decoys, `astralMirage`, and `blind`!

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2. [PATH-OCCULT-02] Path #35: OCCULT_MAGE (วิถีจอมขมังเวทและมหาไสยศาสตร์ — 4 Subpaths)

- **Subpaths**: `YANTRA_IRON_SKIN`, `BLACK_ARTS_HEXER`, `PHRAY_SPIRIT_WHISPER`, `SACRED_GOLEM_TAMER`.
- **Three Live Runtime Bugs in `JS2-v2.6.0`**:
    1. **Missing `'ironSkin'` Entry in `ST.BUFF_CATALOG` (`js/combat.js` vs `js/stats.js:90-145`)**:
        - `COMBAT.COMPOSABLE_ENGINE`'s `KHMER_YANTRA_WARD` action invokes `applyNamedBuff(..., 'ironSkin')`, which is **absent from `ST.BUFF_CATALOG`**!
        - Register `ironSkin` (ยันต์คงกระพันชาตรี — หนังเหนียวฟันแทงไม่เข้า) in `ST.BUFF_CATALOG` & `SK.STATUS`: grants **`+45% DEF` (physical slash/pierce invulnerability), `-10% MDEF` (vulnerable to pure soul/dharma arts), and `poise`**!
    2. **`COMBAT.leave` Prematurely Erases `vowBacklash` (ของเข้าตัว·ผิดสัจจะ) (`js/combat.js`)**:
        - `COMBAT.leave` clears `vowBacklash` the instant a fight ends, removing all out-of-combat consequence of breaking sacred แรงครู precepts! Preserve `vowBacklash` across `COMBAT.leave` until its real `until` timer expires or an `infirmary`/`shrine` purification rite cleanses it!
    3. **Wire Subpath-Authentic Southeast-Asian Occult Mechanics**:
        - `PHRAY_SPIRIT_WHISPER` (พรายกระซิบเตือนภัยแคล้วคลาด): Negates enemy Round-1 ambush/backstab crits (`+30%` Round-1 `eva`) and reduces `explore.js` trap hazards by `-25%`!
        - `SACRED_GOLEM_TAMER` (กุมารทองและควายธนูอาคม): Summons **Kuman Thong** (luck/evasion/omen spirit) or **Khwai Thanu** (`FRONT_ROW` gore-charge talisman bull, immune to biological `poison`/`bleed`)!

- `BLACK_ARTS_HEXER` (คุณไสยมนต์ดำบิดไส้): Armor-bypassing internal hex (`bleed` + `healBlock`), accruing `+sinKarma` and risking `vowBacklash` if reflected by `BUDDHA_DHARMA` / `LOTUS_MERCY_PURIFIER`!

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3. `[PATH-HERETIC-03]` Path #34: `VOID_HERETIC_PACT` (วิถีสัญญานอกรีตและสิ่งมีชีวิตนอกมิติ — 4 Subpaths)

- **Subpaths**: `MADNESS_WHISPERER`, `FLESH_MUTATION_ADEPT`, `VOID_WARPING_HERETIC`, `SANITY_SACRIFICE_WARLOCK`.
- **Verified Bugs & Contextual Contradictions**:
  1. **Unauthorized Sect Treasury Theft (`160` `W.sect.stones`) in Autonomous `formDemonPact` (`js/cultivation.js`)**:
     - Autonomous pact formation deducts `160` stones from **`W.sect.stones`** instead of the warlock's personal `n.inv.stones` (or personal blood/sanity sacrifice)! Require `n.inv.stones >= 160` (crediting `W.world.resourcePool.stones`) or a personal Sanity/Vitality Pact when `W.sect.forbiddenVault.policy !== 'UNRESTRICTED_BLOOD_CULT'`!
  2. **All-Reward / Zero-Madness `sanityStrain` (`stats.js:144`)**:
     - `sanityStrain` grants a massive `+38% matk, +24 pen, +15 crit, +25 critDmg` for only `-10% def`!
     - Honor its `supremeRisk` (**"เผาผลาญสติมากเกินไปเสี่ยงต่อจิตมารและคลุ้มคลั่ง"**): active `sanityStrain` adds `+2.5 heartDemon` per cast; if `heartDemon >= 75`, casts gain `+20%` chaotic Eldritch surge **but** trigger a `15%` chance of **Outer-Void Whisper Leak** (`hallucination` / `-15% acc` for 4 ticks), while `MADNESS_WHISPERER` projects `sanityErosion` (`-willpower / confusion`) onto low-composure foes!

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4. `[PATH-FLESH-04]` Path #36: `DEVOURING_FLESH` (วิถีกลืนกินหลอมสรีระ — 4 Subpaths)

- **Subpaths**: `IRON_BONE_DEVOURER`, `BEAST_FLESH_CONSUMER`, `INTERNAL_ORGAN_REFINER`, `PRIMORDIAL_STOMACH_TITAN`.
- **Two Severe Contextual & Economic Bugs in `js/cultivation.js`**:
  1. **The `|| aliveCults[0]` Fallback Bug (`Orthodox Sect Master Eats Raw Rice & Whetstones!`)**:
     - In `monthlyAutonomousTick`, `const devNpc = aliveCults.find(x => x.cult.path ===`

`'DEVOURING_FLESH') || aliveCults[0]` forces `aliveCults[0]` (typically the **Orthodox `XIAN` Sect Master**!) to devour warehouse items every month even when the sect has **zero** `DEVOURING_FLESH` cultivators! **Delete `|| aliveCults[0]` immediately!**

2. **Unfiltered Warehouse Gluttony & Disconnection From `CONTENT.rareOrgans`**:
    - `devourResourceForBody` blindly consumes the first positive item key in `W.sect.stock`—eating `grain`, `ITEM_REPAIR_STONE`, fishing trash, or priceless Elder breakthrough pills!
    - **Authentic Subpath Diet & Personal Inventory/CP Rule**:
        - Convert into an autonomous `AI.ACTS.devourEssence` action for **living, present (`!n.away && !n.jail`) `DEVOURING_FLESH` cultivators only**, consuming from `n.inv` (or legally requisitioning surplus low-tier stock above safety reserve via CP), strictly filtered by Subpath Diet:
            - `IRON_BONE_DEVOURER` $\to$ **Ores (`ORE`)** $\to$ bounded `ironBoneStacks` (`+def / +poise`).
            - `BEAST_FLESH_CONSUMER` $\to$ **Beast Parts (`BEAST_PART`)** $\to$ bounded `fleshStacks` (`+hp / +atk`).
            - `INTERNAL_ORGAN_REFINER` $\to$ **Mutated Beast Organs (`CONTENT.rareOrgans` / `ORGAN`) & Surplus T1–T3 Pills** $\to$ absorbs organ elemental affinity & `CULT.gain` with `0` organ-rejection risk!
            - `PRIMORDIAL_STOMACH_TITAN` $\to$ Digests any surplus `ORE`/`BEAST_PART`/`HERB` **and** devours enemy shields/summons on `GU_SOUL_LEECH` / `MASS_ABSORPTION` hits in combat!

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5. `[PATH-GRAVE-05]` Path #37: `NETHER_GRAVE` (วิถีสุสานและมารยมโลก — 4 Subpaths)

- **Subpaths**: `ANCESTRAL_GRAVE_KEEPER`, `BONE_PUPPET_REFINER`, `SOUL_SIPHON_REAPER`, `CORPSE_EXPLOSION_ACOLYTE`.
- **Three Verified Code & Contextual Bugs**:
    1. **`W.graveyard` vs Real Sect Death Records & Uncalled `channelGraveyardEcho` (`N-03`)**:
        - `channelGraveyardEcho` had **zero runtime callers** outside `v220` tests, and checks `W.graveyard.length` while real sect/battle deaths are stored in `W.npcs.filter(n => n.dead)` and `W.sect.ancestors`!

    - Implement `SECT.getGraveyardCount(W)` aggregating real fallen graves (`W.graveyard`, non-ascended `W.sect.ancestors`, and fallen `W.npcs`), and wire `AI.ACTS.tendGraveyard` (at `shrine`/cemetery) + pre-raid `boneArmor` (`25% MaxHP`) activation!
2. **Corpse-Count Conservation on `CORPSE_EXPLOSION_ACOLYTE` & `BONE_PUPPET_REFINER`**:
    - `CORPSE_EXPLOSION_ACOLYTE` (สายจุดระเบิดซากศพยมโลก) should **require and consume an actual slain corpse in the current battle (`f._unconsumedCorpses > 0`, incremented in `C.slay`)** to detonate a massive `1.85×` AoE Nether-Blast (falling back to a modest Bone-Shard strike if `0` units have died yet—never exploding imaginary corpses on Turn 1!)!
    - Distinguish **Orthodox Tomb-Wardens (`ANCESTRAL_GRAVE_KEEPER` — honors graves, `+virtueKarma`, `boneArmor` party shield)** from **Necromantic Desecrators (`BONE_PUPPET_REFINER` / `SOUL_SIPHON_REAPER` — `+sinKarma`, prohibited from desecrating allied graves under Orthodox Charters)**!
3. **Nether Yin-Chill Accumulation (**ไอความตายกัดกร่อนความอบอุ่น**)**:
    - Channeling deep grave Yin adds mild `netherChill` fatigue after battle, naturally driving `NETHER_GRAVE` disciples to drink hot tea (`tea` garden) or warm up near the `forge` / `ALCHEMICAL_INCENSE_MIST`!

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6. `[PATH-INTEGRITY-06]` Closing Cross-Cutting Path Hygiene (`N-01`, `N-02`, `N-04`)

To guarantee 100% data and state integrity across all **43 Major Paths and 286 Subpaths**:

2. **Dungeon Prisoner Recruitment (`SECT.handleDungeonPrisoner` in `js/sect.js`)**: When recruiting a prisoner (`RECRUIT_DEFECTOR`), pass `{ path: pris.path }` before rolling subpaths/mantras/techs (or run full path re-initialization: `sub`, `bestMantra`, `initTrack`, `ensurePathGauge`, and purge off-path `techs`) so recruited prisoners never hold 6 off-path techniques!
3. **`bestowPathCrystal` on Existing Cultivators (`js/cultivation.js`)**: Either restrict `CRYSTAL_PATH_*` strictly to mortals (`!npc.cult`), OR invoke `MARTIAL.retrain` logic to cleanly reset `mantra`, `track`, `pathGauge`, `secondaryResources`, and incompatible `techs`!
4. **Resolve All 34 Legacy Orphaned `t.sub` Technique References (`N-02`)**: Add all 34 legacy subpath IDs (`TIDAL_WATER_MIRROR`, `WIND_SHADOW_ROAMER`, `BLOOD_FURNACE`, `THUNDER_PUNISHMENT_ENVOY`, `DIVINE_FORGE_FOUNDER`, `BLOOD_SEA_FIEND`, `SOUL_MUSICIAN`, `ILLUSION_MIND`, `CHANT_RESONANCE`,

GRAND_FORMATION_ARCHITECT, SPIRIT_THREAD_PUPPET, GRAND_STRATEGIST, ASTRAL_LEVIATHAN_KRAKEN, etc.) to CULT.SUB_ALIASES and enforce 0 unresolved t.sub references in tools/validate_data_schema.js!

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7. Master Completion Summary of the 01A–01H Path & Subpath Audit Series

Document	Scope Audited	Paths	Subpaths
01A & 01B	Macro Lifecycle, Path Library, Retraining, Stage Math, Peak Stations & Era-5 Autonomy	All 43	All 286
01C	XIAN, BODY, GOD, DEMON, HEART_SOUL + 9-Realm Track Clone & skillGate Audit	5	65
01D	MECH_PUPPET, KARMA_SAMSARA, BEAST_SOVEREIGN, SAGE_SUPPORT, CRAFT_CREATION, ARRAY_TALISMAN, STRATEGY_GOVERNANCE	7	67
01E	ASTRO_DIVINATION, BUDDHA_DHARMA, VOID_SPACE, CELESTIAL_WEATHER, DRAGON_VEIN_GEOMANCY, GRAVITY_DAO, CHRONO_EPOCH, PRIMORDIAL_CHAOS	8	48

Document	Scope Audited	Paths	Subpaths
01F	MERCANTILE_DAO, INK_LITERATI, LORE_ARCHIVIST, WANDERING_DAO, SONIC_HARMONY, VENOM_GU, SOUL_PUPPETRY, PRIMORDIAL_SHIFTER	8	46
01G	SHADOW_SAMURAI, NINPO_SHADOW, NINPO_ELEMENTAL, ARENA_GLORY, FORTUNE_GAMBLER, TIDE_VOYAGER, SPIRIT_HUSBANDRY, ANCESTRAL_RITUAL	8	32
01H	DIVINE_SUMMONER, DIVINE_POSSESSION, DIVINE_CHANNELER, OCCULT_MAGE, VOID_HERETIC_PACT, DEVOURING_FLESH, NETHER_GRAVE + Cross-Path Hygiene	7	28
TOTAL	**100% Granular Coverage of Every Path & Subpath in `JianghuSect 2`**	**43 / 43**	**286 / 286**